EvidenceVault: Interpretation Profiles, Governed Replay, and Accountable Replay Projection

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Abstract

Portable evidence needs more than bytes, signatures, and hashes: it needs carried meaning, one live governed replay context on the judged scope, and a projection discipline saying which shorter surfaced views remain the same replay claim. This paper is the canonical bridge for that layer: interpretation profiles and admissible translation; governed replay on a fixed scope; and accountable replay projection through replay slices and replay-facing surfaces, including same-scope carrier and presentation consequences on one fixed scope. The bridge consumes only spine-row projections, and family notes declare only local tuples, carriers, and payload forms on admitted same-scope carriers. Silent profile drift, silent policy substitution, stale authority, split-view history splicing, and omission of load-bearing prerequisites therefore cannot preserve a portable ACCEPT.

1 Bridge laws

Interpretation profiles and admissible translation

Definition 1 (Interpretation profile)**.** *An interpretation profile is a tuple*

$$\mathsf{Interp} = (\Xi, \mathsf{Canon}, \rho, \nu, \omega)$$

where:

- Ξ *pins the schema, grammar, or typed object language in which evidence is parsed;*
- Canon *pins the canonicalization and digest rules for structured objects;*
- ρ *assigns semantic roles to carried objects (policy, verifier, waiver, checkpoint, receipt, delegation,* etc.*);*
- ν *pins versioned operational profiles such as transcript ABIs, resolver semantics, anchor rules, roster semantics, and related namespace-specific validity conventions;*
- ω *pins the outcome and composition interface used downstream (for example, what counts as* ACCEPT*, which states are conservative non-accepts, and how leaf receipts compose).*

Profile-bound carried objects. Call a carried object

$$e = (b, d, \pi, \sigma, \mathsf{Interp})$$

profile-bound when b is the object payload, d is its canonical digest under Canon, π is any local proof or receipt about b, and σ is any required authentication. Such an object is portable only relative to the pinned interpretation profile Interp or to a carried translation certificate into another accepted profile.

A digest without an interpretation profile therefore identifies bytes, not meaning: the same byte string can be parsed under the wrong schema, assigned the wrong role, compared under the wrong anchor semantics, mapped to the wrong roster scope, or evaluated under the wrong outcome algebra. This is why EvidenceVault repeatedly binds not only hashes, but also schemas, ABIs, scope coordinates, policy digests, checkpoint rules, and profile identifiers.

Admissible translation versus silent reinterpretation. Portable systems may evolve. The safe way to do that is not silent reinterpretation, but carried translation evidence.

Definition 2 (Translation certificate)**.** *A translation certificate from* Interp_0 *to* Interp_1 *is carried evidence*

$$\tau_{0\to 1}$$

that justifies one of the following under policy:

- *the two profiles are observationally equivalent for the relevant object class;*
- *objects from* Interp_0 *are deterministically re-encoded into* Interp_1 *with preserved semantics; or*
- *replay under* Interp_1 *is explicitly authorized only after revalidation of the affected checks.*

Absent such evidence, treating Interp_0 *objects as if they were native to* Interp_1 *is an unsafe reinterpretation.*

For a family E of profile-bound objects and an accepted profile set P, this same admissible-translation lane first performs the carried-meaning check for portable replay: it validates every object in E under its carried interpretation profile together with any carried translation certificates needed to compare or compose it with objects under another profile. That check owns only carried meaning and admissible cross-profile translation; governed history, authority, temporal, and transparency-obligation checks enter later through policy-aware replay. Missing profile pins or missing translation evidence yield a non-authorizing outcome.

Fail-closed carried-meaning replay therefore treats any change in carried interpretation profile—including the replay-visible operational contracts inside ν, such as transcript ABIs, resolver traces, or anchor-derivation schedules—as a profile change unless the bundle also carries a policy-valid translation certificate for the claimed object class. So a bundle that authorizes replay at this carried-meaning stage cannot preserve acceptance after silently substituting a different carried profile Interp' for Interp unless the two profiles are actually observationally equivalent on that class or an authentication / profile-check primitive is broken. Acceptance therefore depends on carried meaning or carried translation rather than on narration that a new profile is "close enough." For the replay-visible slice of ν, byte-identical replay on every admissible execution under one carried operational-profile label is itself a sufficient equivalence witness.

Governed replay and evolution on one fixed scope

Definition 3 (Portable scope-binding map)**.** *Fix one replay scope. The kernel's judged-scope coordinate stays abstract, but family-local objects still need a carried rule for how their own bundle coordinates, run identifiers, statement identifiers, judged-scope digests inside decision packages, ABOM inventories, and similar names bind to that same scope. Write*

$$B_u = (u, \nu, \chi),$$

for the carried portable scope-binding map on that scope, where u is the canonical scope identity that downstream replay is meant to judge, ν is the namespace/profile material needed to decode family-local identifiers as claims about parts of that scope, and χ is the family-local list of carried component commitments that every such identifier must bind when it claims to speak about u. Typical members of χ are bundle digests, tool or verifier digests, statement identifiers, manifest digests, submission records, policy digests, and carried checkpoint or time-anchor context. The kernel's replay carrier still uses only the judged scope u. The bridge owns only the extra family-local syntax explaining how run IDs, bundle coordinates, judged-scope digests inside decision packages, ABOM inventories, and similar names decode to that same u without permitting silent retargeting.

Proposition 1 (Scope-bound local-name decoding and no-retargeting)**.** *Assume signatures and canonical digests are unforgeable under the pinned profile material in ν. If every supportive, preclusive, and diagnostic object replayed for one decision commits to that same portable scope-binding map B_u, then a purported portable* ACCEPT *for u cannot be replayed as if it justified a different bundle, run, or decision scope without changing some carried scope digest, profile digest, statement identifier, manifest digest, or history/validity evidence named by B_u. Under fail-closed replay, that change is therefore either a different replay claim or a failed authenticity check, not the same portable authorization.*

Proof. If every replayed supportive, preclusive, and diagnostic object binds to the same carried scope-binding map $B_u = (u, \nu, \chi)$, then any attempt to retarget the claim to a different bundle, run, or decision scope must alter at least one carried coordinate named by that map: some scope digest, profile digest, statement identifier, manifest digest, or history/validity commitment in χ. Under the assumed signature and digest soundness for the pinned namespace/profile material in ν, replay therefore sees either a different carried claim or a failed authenticity check rather than the same portable authorization on a new scope. □

Definition 4 (Governed replay context and governed-evolution seam)**.** *A policy-governed judgement is not merely a digest plus a local label. A carried governing policy artifact is a canonical object*

$$P = (\textit{predicate}, \textit{parameters}, \textit{profile}, \textit{authority metadata})$$

with immutable digest

$$h_P = \mathsf{Hash}(\mathsf{Canon}(P)).$$

The carried policy may be embedded directly or referenced by digest through an immutable artifact path. Portable policy-bound meaning is therefore not pinned by carried bytes and profile alone: replay must also fix one bridge-owned governed operativity basis *collecting the coherent history, authority, timing, and transparency state that makes that same policy operative on the judged scope. Write*

$$\mathcal{O} = (\mathcal{H}, \mathcal{A}, \mathcal{W}, \mathcal{T})$$

for that governed operativity basis on the scope, where $\mathcal{H}$ is one coherent accepted history view for every anchor namespace that matters there, $\mathcal{A}$ is the live authority basis on $\mathcal{H}$ for the governance actions that matter there, $\mathcal{W}$ is the conjunction of all temporal predicates derived from carried evidence and policy that must hold at decision time, and $\mathcal{T}$ is the carried transparency-obligation component declaring which externally checkable consistency, publication, observation, or quorum facts are load-bearing for portability on that scope together with the admissible witness forms for those facts. Typical members of $\mathcal{T}$ include inclusion or consistency proofs, checkpoint bindings, witness-quorum receipts, publication-delay bounds, monitor receipts, anti-split-view checks, transcript checkpoints, mutation traces, and comparable externally inspectable observations. Write

$$C = (\mathsf{Interp}, h_P, \mathcal{O})$$

for the carried governed replay context on that scope, where Interp *is the carried interpretation profile and h_P is the canonical digest of the governing policy artifact resolved under* Interp*. A family note may still speak informally of a transparency menu, run-observation surface, monitor menu, or comparable local alias, but unless that alias carries extra local law it is only prose naming for the transparency-obligation component $\mathcal{T}$ inside that same governed operativity basis rather than a second standalone family object. A bare signature therefore supplies only one component of a governed operativity basis rather than the whole replay context, and any history-anchored or authority-relative governance object is portable only relative to one fixed governed replay context.*

Once a policy-bound replay claim is admitted, later evolution is portable only when the change is carried in replayable form on that same governed scope. For scope $\mathcal{S}$, write

$$\varpi = (h_{P_0}, h_{P_1})$$

for the seam's bridge-owned governed policy edge *on that scope, where h_{P_0} and h_{P_1} are the prior and successor policy digests. Also write*

$$\kappa = (\alpha, \eta, \zeta)$$

for the seam's bridge-owned governed control bundle *on that scope, where α is scoped authority evidence for who is allowed to approve that change, η is the anchor-bearing timing evidence that says when the change became live for scope $\mathcal{S}$, and ζ is any carried blocker, supersession, expiry, rollback, waiver, or discharge evidence whose precedence must still replay there. Then write*

$$\gamma = (\varpi, \delta, \kappa)$$

for the bridge-owned governed-evolution seam on that scope, where δ is the canonical change object or translation witness. That tuple γ is the bridge-owned replayable seam for governed evolution on one fixed scope. Call the family-local typed interpretation of its already-carried change-object coordinate δ on one admitted family carrier the seam's local δ-reading*. Family notes therefore do only one thing with γ: they state that local δ-reading and hence only the family-local payload semantics of the already-carried change object on an admitted carrier. Authority, timing, blocker / supersession / rollback / discharge precedence, and any still-load-bearing publication or monitor obligations remain bridge-owned through the governed policy edge ϖ, the governed control bundle κ, the governed replay context, and, when externally checkable, the carried transparency-obligation component $\mathcal{T}$.*

Table 1 therefore records only the current family handoff rows, leaving replay-facing role packages to the admitted carriers that actually define them and leaving seam ownership where it is actually defined rather than repeating either of those as a second architectural surface.

Proposition 2 (Governed operativity-basis completeness on one fixed scope)**.** *If a portable acceptance, waiver, revocation, override, publication claim, or composed receipt depends on a family E of carried external-history witnesses or history-relative authority objects, or on an external-history fact, publication obligation, monitor witness, or transcript-visible observation that is not reproducible from the other carried load-bearing coordinates alone, then replay requires that the governed operativity basis $\mathcal{O}$ inside this carried context be coherent, live, temporally satisfied, and obligation-complete in the following sense:*

1. *the carried governed operativity basis supplies one coherent accepted history view for every anchor namespace invoked by those objects;*
2. *that same governed operativity basis supplies one live authority basis for every signed governance, roster, delegation, or approval object whose authorization is decision-relevant on that scope;*
3. *that same governed operativity basis carries one temporal window $\mathcal{W}$ whose required predicates hold at decision time; and*
4. *each non-redundant external-history fact, publication obligation, monitor witness, or transcript-visible observation appears on the carried transparency-obligation component $\mathcal{T}$ either as carried evidence or as a carried commitment whose witness form is declared there and later replay can resolve on the same scope.*

For a family E of profile-bound objects and an accepted profile set $PSet$, write

$$\mathsf{ProfileEval}(E, PSet) \in \{\mathsf{ACCEPT}, \mathsf{REJECT}, \mathsf{UNKNOWN}\}$$

for the governed refinement of that carried-meaning check on that same scope.

Proof. The carried governed replay context $C = (\mathsf{Interp}, h_P, \mathcal{O})$ fixes exactly one governed operativity basis $\mathcal{O} = (\mathcal{H}, \mathcal{A}, \mathcal{W}, \mathcal{T})$ whose four coordinates portable replay must recover on that scope. If carried external-history witnesses for the same namespace sit on incomparable checkpoints without carried reconciliation evidence, then $\mathcal{H}$ is not one coherent accepted history view. If a signed governance object lacks a live authority chain on that coherent view, then $\mathcal{A}$ is not one live authority basis for the claimed action, scope, or namespace at decision time. If a required freshness, not-before, expiry, publication-delay, or renewal guard fails, then $\mathcal{W}$ is not satisfied at decision time. And if a needed consistency proof, publication witness, monitor receipt, checkpoint, or transcript-adjacent observation is neither carried nor commitment-addressed on $\mathcal{T}$, then downstream replay cannot distinguish "this fact was discharged" from "this fact was never exposed for checking." So fail-closed portability cannot authorize reuse from silent history splicing, unauthenticated authority claims, expired governance windows, or hidden external obligations; the carried governed operativity basis must therefore satisfy all four conditions together. □

Definition 5 (Policy-target witness)**.** *Fix one governed replay context C and one policy target*

$$\xi = (h_x, h_P),$$

where h_x is the judged object digest and h_P is the governing policy digest from that same context. A policy-target witness *for ξ on C is carried evidence that is either*

(a) a replayable evaluator, replay-slice surface, or decision transcript that lets a verifier recompute whether $P(x) = 1$ on that same policy target ξ, or

(b) a sound proof π of the statement "$P(x) = 1$", where the statement commits to that same policy target ξ.

Such a witness is portable only when it is itself scoped to that same policy target ξ on that same governed replay context C.

Proposition 3 (Policy-target replay binding on one fixed scope)**.** *Beyond the shared governed replay context C, policy-aware replay additionally requires only that every policy-relative acceptance branch bind one carried* policy target

$$\xi = (h_x, h_P),$$

and that:

1. *the policy coordinate of ξ resolves under its carried profile to one immutable governing policy artifact with that same canonical digest; and*

2. *whenever some carried acceptance branch asserts a policy-relative outcome for that same policy target ξ, the associated claim or receipt commits to that same policy target and additionally carries one policy-target witness for ξ on that same governed replay context C.*

The bridge treats ξ itself as the portable object–policy boundary for that branch on that scope. Carrying the policy coordinate of ξ alone preserves interpretability but not portable acceptance; carrying a policy-target witness without that same bound policy target preserves neither meaning nor replay. Mixing incomparable checkpoints, stale roster views, expired validity windows, undeclared external obligations, or silently substituted policy targets cannot preserve a portable ACCEPT*.*

Proof. Given that same context, portable acceptance for a policy-governed object is also specific to one policy target $\xi = (h_x, h_P)$. If the policy coordinate of ξ does not resolve to one immutable policy artifact under the carried profile, then replay lacks a stable governing predicate. If a claimed acceptance branch omits the policy target or scopes its policy-target witness to a different target or governed replay context, then replay no longer verifies that the carried policy accepted the carried object on that same governed context. And if checkpoint, roster, delegation, or timing material is silently substituted by incomparable, stale, lapsed, or expired material without the carried reconciliation or renewal evidence that policy requires, then the governed replay context has changed rather than the same portable claim being replayed. Thus fail-closed replay treats omission or silent substitution of policy-target identity, policy-target-witness binding, or governed-context coordinates as non-authorizing unless carried translation, ratchet, reconciliation, or renewal evidence restores that same replay claim. □

Definition 6 (Conservative ratchet witness)**.** *Fix one governed seam γ on one fixed governed scope. Write*

$$\vartheta = (\varpi, \delta)$$

for its bridge-owned conservative ratchet witness *on that scope when the carried interpretation profile reads both policy artifacts named by ϖ as conjunctions of checks over the same admitted carrier family there, and the carried change witness δ certifies that every predecessor check either remains unchanged in the successor or is replaced there by a monotone stronger check with the same scope and semantics while any genuinely new successor clause is only an additional required check on that same scope. The bridge uses ϑ exactly for that monotone predecessor/successor policy reading, keeping ρ reserved for the interpretation profile while leaving replay-bearing authority, timing, blocker, supersession, rollback, waiver, and discharge coordinates on κ.*

Proposition 4 (Conservative governed evolution on one fixed scope)**.** *On that same carried seam, conservative conjunctive ratchets are strengthenings rather than fresh local acceptance doctrines: if one governed seam γ carries a conservative ratchet witness ϑ on that fixed scope, then every object accepted under the successor policy on that scope would also have been accepted under the predecessor policy there. So such a ratchet is conservative on that same live governed scope, with the replay assumptions on κ keeping that implication from being manufactured by silent authority, timing, or blocker substitution.*

Proof. Now fix one replay scope, one carried governed replay context on that scope, and one governed seam γ that replays successfully there. If ϑ is a conservative ratchet witness on that same scope, then by definition the carried interpretation profile reads the successor policy as a monotone strengthening of the predecessor on the same admitted carrier family there. So every object accepted under the successor policy on that scope must satisfy every predecessor clause of the predecessor policy there as well. Therefore acceptance under the successor policy implies acceptance under the predecessor policy on that same governed scope, and the replay-bearing governed seam γ keeps that implication from being manufactured by silent policy, authority, timing, or blocker substitution. □

Accountable replay projection and replay-facing surface discipline

Definition 7 (Accountable replay projection)**.** *Fix one governed replay state on one decision or policy scope. Accountable replay projection is the bridge's rule for carrying a shorter visible view of that state on that fixed scope. Relative to a claim and replay policy, call an object* load-bearing *when substituting, deleting, or mutating it can change a downstream replay outcome; typical examples are policy texts, verifier identities, canonicalization rules, transparency checkpoints, retry budgets, schema versions, and signed receipts. An accountable replay projection is any carried surface that chooses a visible projection while keeping every omitted load-bearing coordinate live through carried commitments, digest/role bindings, the governing interpretation profile, and residual obligations sufficient for fail-closed replay on that same scope. Accordingly, when replay sees an omitted object only through its canonical digest under that governing interpretation profile, any substitute with the same digest under the same canonicalization rule is replay-equivalent there. Decision-local replay slices are the canonical mechanism-shaped instance of this same rule.*

Proposition 5 (Omission-conservative replay from accountable projection)**.** *Fix one fuller replay state E on one decision or policy scope. If* Replay *is fail-closed and information-monotone in the direct sense that replacing load-bearing coordinates only by explicit unknown markers, commitments / hashes, or carried obligations never turns a non-authorizing fuller state into an authorizing projected one by default, then any accountable replay projection* Proj *on that same scope satisfies*

$$\mathsf{Replay}(\mathsf{Proj}(E)) = \mathsf{ACCEPT} \Longrightarrow \mathsf{Replay}(E) = \mathsf{ACCEPT}.$$

Thus acceptance on the shorter surfaced view can witness acceptance on the fuller replay state only when every omitted load-bearing coordinate stays carried, commitment-addressed, or discharged under the same interpretation profile and residual obligations; a projection may forget, but it may not improvise*.*

Definition 8 (Replay slices)**.** *A* replay slice *for evidence state E under policy π is a tuple*

$$\Sigma = (\mathsf{Proj}(E), H, M, \Pi, \Omega),$$

where $\mathsf{Proj}(E)$ *is the visible projected evidence, H is a commitment set for omitted load-bearing coordinates, M is a manifest that maps visible and omitted load-bearing objects to their canonical*

digests and roles, Π *is the interpretation/replay profile needed to decode the slice and check it against the fixed governed replay context, and* Ω *is the finite carried family of residual obligations that keeps unresolved prerequisites live downstream. A replay slice is well-formed only when every omitted load-bearing coordinate for the fixed decision or policy scope is either (i) recorded by commitment or digest in* H *and* M*, (ii) reintroduced as an element of* Ω*, or (iii) justified as irrelevant to that scope by the carried policy/profile material in* Π*. So scope-local completeness and residual-obligation carriage are internal to the slice form rather than neighboring bridge objects. The bridge owns that replay-slice object model once: the spine and family notes may carry a replay slice* Σ *abstractly and cite its consequences, but they do not re-define supportive/preclusive membership, omission discipline, or surfaced-view variants beside it.*

Definition 9 (Replay-slice surfaces and row handoff)**.** *Fix one replay slice* Σ *on one governed replay context. A* replay-slice surface *is a carried tuple*

$$S = (\phi, V, \upsilon),$$

where ϕ *is a surface kind,* V *is the visible presentation, and* υ *carries any extra authentication or retrieval-discharge data for that presentation, while every omitted load-bearing coordinate remains live through the commitments* H*, manifest* M*, interpretation profile* Π*, and residual obligations* Ω *of the underlying slice. The bridge uses the following canonical kinds:*

1. paper*:* $V = (P, D, R)$ *where* P *is paper text,* D *is the set of canonical digests for any load-bearing objects not embedded in* P*, and* R *is the retrieval rule explaining where the omitted objects live and what replay guarantee is available before retrieval;*

2. front*:* $V = (v, \iota)$ *where* v *is a compact vector of policy-relevant claims extracted from* $\mathsf{Proj}(E)$ *and* ι *is a digest-addressed source map into the manifest* M *for the load-bearing objects that justify those claims, while* $\upsilon = \sigma$ *authenticates the canonical encoding of* (v, ι)*; and*

3. closure*: any replay-slice surface whose retrieval rule has been discharged far enough that offline replay no longer depends on mutable network pointers or ambient infrastructure.*

For cross-family bookkeeping on one fixed scope, the bridge also uses two optional row labels, F *and* C*, where* F *is either* $\perp$ *or a carried conservative role-escalation / front-full comparison surface on that scope and* C *is either* $\perp$ *or a citation-role surface on that same slice. So* C *does not introduce a second citation doctrine beside replay-facing role packages: it is only the row-level label for a domain-local handle on the replay slice together with the decision or scope identity that the slice is meant to justify, while family-specific headers, display labels, or compact front matter stay presentation-only unless the carried replay-facing role package says otherwise. A separately surfaced lineage export likewise needs no third row label: on one fixed slice it is judged only through the same replay-facing surface discipline, together with whether the underlying lineage fact is already carried or otherwise discharged on that scope. The spine owns the normalized row declaration* (R, K, F, C)*. Call its direct bridge-side record a* family handoff row*: it records only the surviving scope-fixing local form* R*, admitted carrier family* K*, and instantiated bridge labels* F *and* C *at that interface. That replay-slice surface taxonomy is also the row-handoff rule: after the spine declares a row, later bridge or family prose cites only the members of the family handoff row it actually consumes, while any replay-facing role package is still read only where an admitted carrier actually defines it.*

Replay-facing frame. For the next replay-facing bridge surfaces, fix one replay scope, one admissible row-local carrier reading K on that scope, and one replay slice on that scope. Those surfaces all speak relative to that same fixed frame unless they say otherwise.

Definition 10 (Replay-facing surfaces and role packages)**.** *A* replay-facing surface *for that slice is any accountable replay projection or scope-fixed wrapper of that slice that a producer presents downstream on that scope as a fast path, full replay view, citation view, lineage export, admission surface, audit surface, or comparable replay-facing role. For any replay-facing surface, write*

$$\psi = (k, \mathcal{B})$$

for its replay-facing role package*, where k is a deterministic replay-facing role selector and $\mathcal{B}$ is the finite family of load-bearing role-binding coordinates that downstream replay must inspect in order to interpret k on that scope. Typical members of $\mathcal{B}$ are carried conformance suites, differential witnesses, profile or policy pins, role-specific residual obligations, and any scoped authority or lineage / invariance material without which the claimed role would be ambiguous or overstate what the surface actually justifies. The bridge treats that pair ψ itself as the portable role-boundary object for that surface on that scope. Replay-facing role claims therefore live in the bridge-owned surface taxonomy rather than in row-local carrier identity: a bare digest, URL, prose reference, or nominal citation handle is not a portable replay boundary unless it pins the relevant bytes and semantics on that scope, and any narrower policy-shaped front, audit, citation, lineage, or comparable surface is judged only as a claimed replay-facing role on that same slice.*

Proposition 6 (Replay-facing role-package completeness)**.** *Call a replay-facing surface* role-package-complete *exactly when it either carries or commitment-addresses its full replay-facing role package ψ. Portable downstream reuse may rely on its claimed role only if the surface is role-package-complete. If the selector k or any load-bearing member of $\mathcal{B}$ is omitted without commitment, the safe replay result is downgrade, explicit residual obligation, or failure rather than silent acceptance of the role label, because replay otherwise cannot tell whether the visible surface merely summarized the same fixed slice or silently omitted a role-bearing prerequisite.*

Proposition 7 (Conservative narrower/fuller-role reuse on one fixed slice)**.** *If a claimed narrower replay-facing surface is supposed to be authorizing only because it is conservative relative to a fuller replay-facing role on that same fixed slice, then portable downstream reuse may rely on the narrower role only when either policy proves that no fuller role is needed on that scope or the bundle carries a conservative-comparison tuple*

$$\epsilon = (k_f, k_a, D_f, D_a, c),$$

where k_f and k_a are the narrower and fuller replay-facing role selectors on that slice, D_f and D_a are their carried visible surfaces, and c is replayable relation evidence showing that both surfaces speak about the same judged scope and replayable history / validity basis, that the narrower surface is a conservative accountable replay projection of the fuller one on that fixed slice, and that any producer-chosen branch path, escalation path, verifier menu, stopping point, or surfaced projection affecting that comparison is either itself carried or already discharged as branch-invariant on that same scope. Without either that carried conservative-comparison tuple or policy evidence that no fuller role is needed, fail-closed replay cannot authorize reuse of the narrower role.

Definition 11 (Row-local representations, replay basis, presentation shells, replay handles, and presentation-only differences)**.** *Within that same replay-facing frame, call a* row-local representation *either that carrier together with its typed payload roles or direct tag/payload fields, or a scope-fixed wrapper over one or more such carriers that batches, projects, or presents them—for example through local headers, display labels, fetch hints, transport metadata, or submission containers—without changing commitments, profiles, omission commitments, blocker handling, or residual obligations. For that carried reading, write*

$$B_K$$

for its replay basis *on that scope: the finite family of replay-bearing coordinates whose preservation keeps the same carrier reading there. Concretely, B_K contains the replay-bearing commitments, interpretation profile, omission commitments, blocker handling, residual obligations, and any other carried coordinates that portable replay must preserve in order to keep the same carrier reading on that fixed scope. Write*

$$S_K$$

for its presentation shell *on that scope: the family of batching, support indexing, local headers, display labels, fetch hints, transport or submission containers, byte-preserving storage forms, producer-side emitter / coordinator / operator aliases, citation handles, replay-facing role labels, and comparable outer surfaces attached outside the preserved replay basis B_K. Support manifests and comparable multi-object support handles are therefore not wrappers merely because they aggregate several package digests: when local replay expands them directly on one fixed scope, they are coequal row-local carriers rather than merely part of the presentation shell S_K. Call any canonical digest, authenticated name, signed header, detached signature target, or comparable canonical handle computed for portable replay on that scope a* replay handle*; it is* carrier-identifying *when it authenticates exactly the replay basis B_K and* surface-identifying *when it authenticates one chosen replay-facing surface of that same carrier. Call a difference between two admissible row-local representations of that same carried reading* presentation-only *when it changes only the presentation shell S_K while preserving the replay basis B_K.*

Proposition 8 (Same-claim presentation on one fixed slice)**.** *Fix two admissible row-local representations R_1 and R_2 of that same carried reading within the same replay-facing frame. If R_1 and R_2 preserve the same replay basis B_K of that carried reading, and every carried digest-addressed or signed component shared between them has the same canonical bytes under the interpretation profile's canonicalization rule* Canon*, then downstream replay reads the same portable claim on that scope from R_1 and R_2. Presentation-only differences therefore cannot strengthen or weaken portability: any replay handle $a(R)$ is carrier-identifying exactly when its covered bytes are the replay basis B_K and otherwise only surface-identifying for the chosen presentation R, an authenticated local name that also functions as a replay-facing role selector still requires the full replay-facing role package required by the preceding completeness proposition, and byte-preserving transport or labeling choices inside the presentation shell S_K are same-claim presentation refinements rather than stronger portable claims. Changing one coordinate of B_K changes the carried replay claim rather than merely repackaging it, while changing the canonical bytes of a carried digest-addressed or signed component either changes the named carried object or causes the corresponding authenticity check to fail.*

Family	R scope-fixing local form	K admitted local carriers	F	C
StreamFold	σ_{run}.	M_{run}; RV.	✓	✓
zk-RTP	Φ.	E; SM.	$\perp$	✓
PACT / OCF	(u, θ) inline in Pkg.	Pkg.	✓	$\perp$

Table 1: Bridge-owned family architecture map for the current mechanism notes. Read each row as one family handoff row of (R, K, F, C), so families carry scope-fixing coordinates and any replay-facing role package only where the admitted carriers themselves define it, while governed-replay seam ownership stays with the governed-replay propositions above and same-scope consequences are derived by the preceding replay-facing role-package / narrower-fuller comparison propositions and same-claim proposition.